

Answer Key — Physics Grade 10

Sub-Strand 2.1: Properties of Waves • Lesson 1: Anchoring Phenomenon: Echoes at Hell's Gate, Ngong, Kisumu and Nairobi

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	B	D	D	D	B	B

USE AFTER: PREDICT PHASE — Predict: individual notebook prediction 'What do you think an echo is?'

Q1. Before any teaching, Wanjiru writes down what she thinks an echo is. Which of these is the best everyday description of an echo?

- A. A sound that stays near your mouth and gets louder
- B. A brand new sound that the wall makes by itself
- C. Your own sound coming back to you after it meets a surface**
- D. A sound that only the wind can make

Correct answer: C. At this stage an echo is described simply as sound coming back after meeting a surface. The idea that the wall makes a new sound is a common misconception to be tested later.

USE AFTER: OBSERVE PHASE — Part B: studying the Echo Table for the four trip sites (A and B)

Q2. The Echo Table shows a clear echo after 0.50 s at the Hell's Gate cliff (85 m away) and a clear echo after 1.00 s at the Ngong Hills valley wall (170 m away). Which statement is supported by these observations?

- A. The delay was the same at both places
- B. The echo at Ngong Hills came back after a longer delay than at Hell's Gate**
- C. The echo at Ngong Hills came back after a shorter delay than at Hell's Gate
- D. No echo was heard at Ngong Hills

Correct answer: B. The table gives 1.00 s at Ngong and 0.50 s at Hell's Gate, so the Ngong delay was longer. This is only a description of the data, with no calculation needed.

USE AFTER: OBSERVE PHASE — Part B: Echo Table, site C (Kisumu school hall)

Q3. In the Kisumu school hall, the far wall is 12 m away and the delay is 0.07 s. What did the learners record about the sound they heard there?

- A. A faint, clear echo after 0.50 s
- B. No echo at all
- C. A clear echo after 1.00 s
- D. A loud, muddy and drawn-out sound**

Correct answer: D. The Echo Table records site C as loud, muddy and drawn-out. This differs from the clear echoes at the two outdoor cliff and valley sites.

USE AFTER: OBSERVE PHASE — Part B: Echo Table 'What I notice' column, comparing sites A, B, C and D

Q4. Learners recorded what they heard at four sites: Hell's Gate cliff, Ngong Hills valley wall, Kisumu school hall and Nairobi school open grounds. Which two sites gave a clear echo?

- A. Kisumu school hall and Nairobi school open grounds
- B. Ngong Hills valley wall and Kisumu school hall
- C. Hell's Gate cliff and Nairobi school open grounds
- D. Hell's Gate cliff and Ngong Hills valley wall**

Correct answer: D. Hell's Gate and Ngong both gave a clear echo. Kisumu was muddy and drawn-out, and Nairobi gave no echo.

USE AFTER: EXPLAIN PHASE — Explain: pair talk, prompt 3 'Why might the Kisumu hall sound muddy and the Nairobi field give no echo at all?'

Q5. In the Echo Table, the Nairobi school open grounds had no hard wall nearer than 300 m and no echo was heard. At this point in the unit, which is the most honest answer to 'Why was no echo heard there?'

- A. Sound disappears when there are no walls close by
- B. The grass swallows all the sound, so nothing can return
- C. The wind always carries the sound away on open ground
- D. We have only recorded what was heard; the reason is still a question we will investigate**

Correct answer: D. This is the anchor lesson, so learners have only observed and predicted. The other options state guesses as facts, and they are ideas the class will test in later lessons.

USE AFTER: DQB PHASE — DQB: learners write 2 or 3 'Why' or 'How' questions on sticky notes

Q6. The class is building a Driving Question Board about echoes. Which learner question is best for the board because it asks about a cause?

- A. Who wrote the story about the four learners?
- B. Why is there a delay before the echo at Hell's Gate is heard?**
- C. In which county is Hell's Gate National Park?
- D. How many learners went on the trip to Hell's Gate?

Correct answer: B. Good Driving Question Board questions start with Why or How and ask about causes of the phenomenon. The other options are simple facts that do not help explain echoes.

USE AFTER: MODEL BUILDING PHASE — Model: drawing the Initial Echo Model with arrows and short labels

Q7. Amina is drawing her Initial Echo Model of a shout at a cliff. Which set of items should her drawing include?

- A. Only the cliff, because the cliff is the only thing that matters
- B. The person shouting, the cliff, and arrows with labels showing what she thinks travels to the cliff and comes back**
- C. Only the person shouting, because the sound stays with the speaker
- D. A final answer with a formula, so that nobody can say the model is wrong

Correct answer: B. The first model should show a source, a surface and a returning path with labels. It is an initial idea that will be revised, so it need not be fully correct, and no formula is used yet.